Experiment Report: Study of Polarisation of Light

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1 Objectives

- To analyse linearly polarized light.
- To verify Malus's law.
- To rotate the state of polarization of a linearly polarized light using a half-wave plate.
- Conversion of linearly polarized light into circularly polarized light using quarter-wave plate.

2 Working Formula

The polarisation of an electromagnetic wave is the information of the direction of oscillation of the electric and magnetic field of the wave.

Light with the electric field oscillating in only one direction is called linearly/plane polarised light with the direction of oscillation being the polarisation direction. One of the ways to create plane polarised light is to use a polarising filter that blocks the component of the electric field in a particular direction.

Polarised light can be obtained from unpolarised light by using birefringent materials (e.g. calcite $CaCO_3$ crystals) which shows anisotropy in their refractive index. The refractive index of such materials depends on the direction of polarisation of light passing through it. One component of the light falling on such a material travels with the same velocity in every direction of the crystal is known as the the ordinary ray. The other component of light ray whose velocity depends on the direction of propagation is called the extraordinary ray. The refractive indices of the ordinary and extraordinary ray are n_o and n_e respectively.

2.1 Half-wave plate and quarter wave plate

A half-wave plate is an optical device that causes a phase difference of π to one of the components of the light ray passing through it as compared to the other component. The axis along which light has the higher velocity is the fast axis of the wave plate. The component oriented along the slow axis gets shifted by a phase difference of π as compared to the component oriented along the fast axis. An incident light oriented at an angle of θ with respect to the fast axis gets rotated by 2θ in its polarisation over the fast axis.

A quarter wave plate is an optical device that causes a phase difference of $\pi/2$ to one of the components of the light ray passing through it as compared to the other component.

2.2 Malus's law

When a plane polarised light is incident on the analyser, the intensity of the light that emerges on the other side depends on the orientation of the analyser with respect to the polarisation angle of the plane polarised light. If the angle of orientation of the analyser is θ with respect to the polarisation angle, the intensity I of the emerging light is given by:

$$I = I_0 \cos^2 \theta \tag{1}$$

where I_0 is the maximum intensity (or voltage reading) of the polarised light. Hence, we expect a linear relationship between $\cos^2 \theta$ and I.

3 Observations

Table 1: Analysis of linearly polarised light

Angular position of polariser = 62°

Sl No.	Analyser angle (θ')(deg)	Corrected angle (θ)(deg)	V (mV)	Sl No.	Analyser angle (θ')(deg)	Corrected angle (θ)(deg)	V (mV)
1	255	186	455	22	60	351	453
2	260	191	454	23	70	1	455
3	270	201	449	24	80	11	452
4	290	221	425	25	90	21	447
5	310	241	319	26	110	41	410
6	315	246	234	27	130	61	264
7	320	251	164	28	120	51	380
8	325	256	100	29	135	66	180
9	330	261	50	30	140	71	113
10	335	266	15	31	145	76	61
11	340	271	0	32	150	81	22
12	350	281	28	33	155	86	0
13	349	280	18	34	160	91	1
14	345	276	5	35	165	96	24
15	355	286	90	36	170	101	59
16	0	291	132	37	180	111	185
17	5	296	235	38	190	121	346
18	10	301	302	39	220	151	443
19	20	311	384	40	210	141	428
20	30	321	418	41	250	181	455
21	50	341	447	–	–	–	–

Table 2: Verification of Malus's law (data obtained from table 1)

Angular position of polariser = 62°

Sl No.	Corrected angle (θ)(deg)	$\cos^2\theta$	V (mV)	Sl No.	Corrected angle (θ)(deg)	$\cos^2\theta$	V (mV)
1	186	0.99	455	22	351	0.98	453
2	191	0.96	454	23	1	1.00	455
3	201	0.87	449	24	11	0.96	452
4	221	0.57	425	25	21	0.87	447
5	241	0.24	319	26	41	0.57	410
6	246	0.17	234	27	61	0.24	264
7	251	0.11	164	28	51	0.40	380
8	256	0.06	100	29	66	0.17	180
9	261	0.02	50	30	71	0.11	113
10	266	0.00	15	31	76	0.06	61
11	271	0.00	0	32	81	0.02	22
12	281	0.04	28	33	86	0.00	0
13	280	0.03	18	34	91	0.00	1
14	276	0.01	5	35	96	0.01	24
15	286	0.08	90	36	101	0.04	59
16	291	0.13	132	37	111	0.13	185
17	296	0.19	235	38	121	0.27	346
18	301	0.27	302	39	151	0.76	443
19	311	0.43	384	40	141	0.60	428
20	321	0.60	418	41	181	1.00	455
21	341	0.89	447	–	–	–	–

Table 3: Rotation of polarisation using half-wave plate

Angular position of polariser = 62°
Max intensity position of half-wave plate = 247° Angular position of half-wave plate = 292°

Sl No.	Analyser angle (θ')(deg)	Corrected angle (θ)(deg)	V (mV)	Sl No.	Analyser angle (θ')(deg)	Corrected angle (θ)(deg)	V (mV)
1	250	181	0	23	60	351	22
2	255	186	6	24	65	356	0
3	260	191	34	25	70	1	1
4	265	196	82	26	75	6	23
5	270	201	143	27	80	11	69
6	275	206	231	28	85	16	127
7	280	211	305	29	90	21	198
8	285	216	359	30	95	26	284
9	290	221	386	31	100	31	343
10	295	226	406	32	120	51	426
11	300	231	419	33	110	41	400
12	320	251	447	34	140	71	449
13	340	271	453	35	160	91	453
14	0	291	445	36	170	101	449
15	20	311	415	37	190	121	426
16	25	316	399	38	200	131	399
17	30	321	374	39	210	141	342
18	35	326	328	40	220	151	195
19	40	331	269	41	215	146	280
20	45	336	173	42	230	161	57
21	50	341	115	43	240	171	2
22	55	346	56	44	225	156	113

Table 4: Circularly polarised light using quarter-wave plate

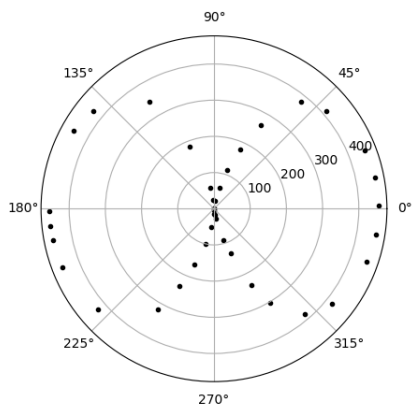
Angular position of polariser = 181°
Angular position of quarter-wave plate = 226°

Sl No.	Analyser angle (θ')(deg)	Corrected angle (θ)(deg)	V (mV)
1	0	179	388
2	20	199	391
3	40	219	394
4	60	239	396
5	80	259	396
6	100	279	394
7	120	299	391
8	140	319	388
9	160	339	386
10	180	359	387
11	200	19	390
12	220	39	394
13	240	59	395
14	260	79	396
15	280	99	394
16	300	119	391
17	320	139	388
18	340	159	387

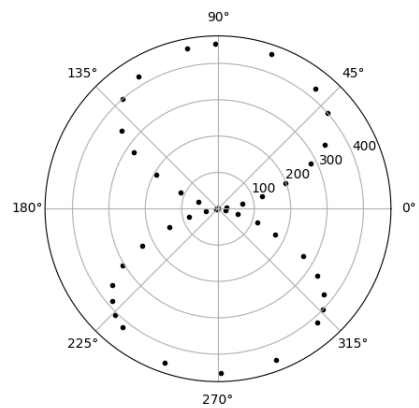
Table 5: New data for verifying Malus's law

Resistance R = 460 Ω
Angular position of polariser = 12°

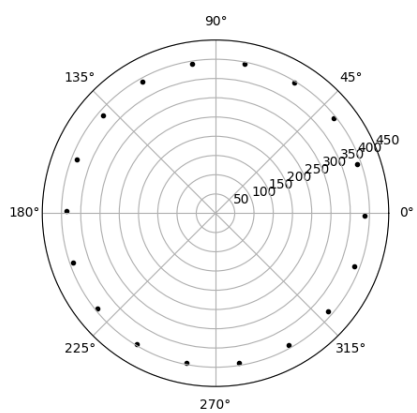
Sl No.	Analyser angle (θ') (deg)	Corrected angle (θ) (deg)	$\cos^2\theta$	V (mV)	Sl No.	Analyser angle (θ') (deg)	Corrected angle (θ) (deg)	$\cos^2\theta$	V (mV)
1	0	6	0.99	746	10	180	186	0.99	755
2	20	26	0.81	713	11	200	206	0.81	716
3	40	46	0.48	556	12	220	226	0.48	545
4	60	66	0.17	217	13	240	246	0.17	199
5	80	86	0.00	11	14	260	266	0.00	10
6	100	106	0.08	60	15	280	286	0.08	71
7	120	126	0.35	341	16	300	306	0.35	351
8	140	146	0.69	651	17	320	326	0.69	656
9	160	166	0.94	740	18	340	346	0.94	735



(a) Only polariser and analyser

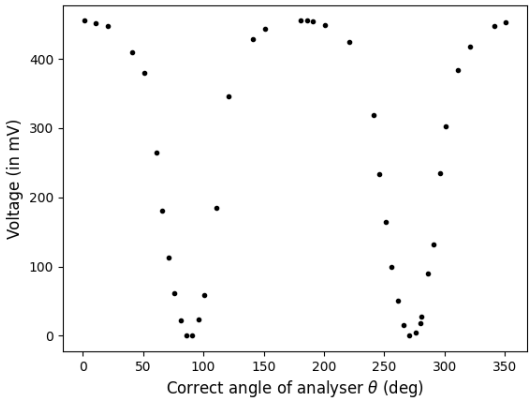


(b) With the half-wave plate

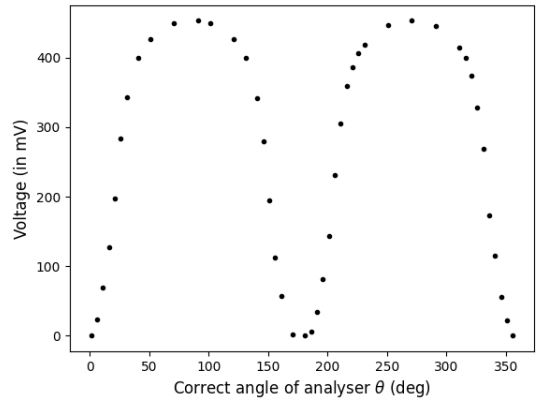


(c) With the quarter-wave plate

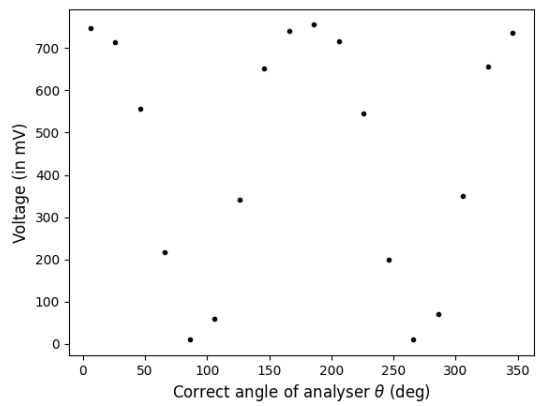
Figure 1: Polar plots of the voltage reading with corrected angular position of analyser from tables 1, 3, 4



(a) Only polariser and analyser (table 1)



(b) With the half-wave plate (table 3)



(c) Only polariser and analyser (table 5)

Figure 2: Graph plots of the voltage reading with corrected angular position of analyser from tables 1, 3, 5

4 Calculations and error analysis

A least square linear fit is performed for the data in table 2 and table 5. In table 6 and table 7, the x_i are $\cos^2\theta$ and y_i are V of table 2 and table 5 respectively.
Let $y_i = A + Bx_i$ where A and B are the parameters such that the mean square error, defined as:

$$\chi^2 = \sum_i \frac{(y_i - A - Bx_i)^2}{\sigma_{y_i}^2}$$

is minimised. From the theory of least square linear fit, we know that the required values of A and B with their errors is given by:

$$\Delta = N\sum x_i^2 - \left(\sum x_i\right)^2 \qquad \sigma_y = \sqrt{\frac{1}{N-2}\sum_{i=1}^{N}(y_i - A - Bx_i)^2}$$

$$A = \frac{\sum x_i^2 \sum y_i - \sum x_i \sum x_i y_i}{\Delta} \qquad \sigma_A = \sigma_y\sqrt{\frac{\sum x_i^2}{\Delta}}$$

$$B = \frac{N\sum x_i y_i - \sum x_i \sum y_i}{\Delta} \qquad \sigma_B = \sigma_y\sqrt{\frac{N}{\Delta}}$$

Table 6: Least square linear fitting for Malus's law using table 2 data

Sl No.	x_i ($\cos^2\theta$)	y_i (V) (mV)	x_i^2	x_iy_i (mV)	$\left(y_i - A - Bx_i\right)^2$ (mV^2)	Sl No.	x_i $\cos^2\theta$	y_i V (mV)	x_i^2	x_iy_i (mV)	$\left(y_i - A - Bx_i\right)^2$ (mV^2)
1	0.99	455	0.98	450.03	4265.68	21	0.89	447	0.80	399.62	963.78
2	0.96	454	0.93	437.47	3023.13	22	0.98	453	0.95	441.91	3756.45
3	0.87	449	0.76	391.34	363.71	23	1.00	455	1.00	454.86	4904.84
4	0.57	425	0.32	242.07	8315.95	24	0.96	452	0.93	435.54	3247.06
5	0.24	319	0.06	74.98	17937.53	25	0.87	447	0.76	389.59	443.99
6	0.17	234	0.03	38.71	6380.45	26	0.57	410	0.32	233.53	5805.19
7	0.11	164	0.01	17.38	1318.04	27	0.24	264	0.06	62.05	6230.11
8	0.06	100	0.00	5.85	43.44	28	0.40	380	0.16	150.50	15214.92
9	0.02	50	0.00	1.22	1718.11	29	0.17	180	0.03	29.78	669.66
10	0.00	15	0.00	0.07	4587.81	30	0.11	113	0.01	11.98	215.95
11	0.00	0	0.00	0.00	6513.36	31	0.06	61	0.00	3.57	2078.52
12	0.04	28	0.00	1.02	4727.53	32	0.02	22	0.00	0.54	4823.32
13	0.03	18	0.00	0.54	5772.40	33	0.00	0	0.00	0.00	6844.82
14	0.01	5	0.00	0.05	6468.63	34	0.00	1	0.00	0.00	6352.95
15	0.08	90	0.01	6.84	592.87	35	0.01	24	0.00	0.26	3773.37
16	0.13	132	0.02	16.95	32.14	36	0.04	59	0.00	2.15	1425.59
17	0.19	235	0.04	45.16	4759.83	37	0.13	185	0.02	23.76	2240.23
18	0.27	302	0.07	80.11	10710.91	38	0.27	346	0.07	91.78	21754.34
19	0.43	384	0.19	165.28	12559.28	39	0.76	443	0.59	338.88	498.58
20	0.60	418	0.36	252.45	4748.76	40	0.60	428	0.36	258.49	6226.98
–	–	–	–	–	–	41	1.00	455	1.00	454.86	4904.84
–	–	–	–	–	–	Total	14.84	9902.00	10.83	6011.20	207215.06

Table 7: Least square linear fitting for Malus's law using table 5 data

Sl No.	x_i ($\cos^2\theta$)	y_i (V) (mV)	x_i^2	x_iy_i (mV)	$\left(y_i - A - Bx_i\right)^2$ (mV^2)	Sl No.	x_i $\cos^2\theta$	y_i V (mV)	x_i^2	x_iy_i (mV)	$\left(y_i - A - Bx_i\right)^2$ (mV^2)
1	0.99	746	0.98	737.85	6470.16	10	0.99	755	0.98	746.75	5103.29
2	0.81	713	0.65	575.98	708.64	11	0.81	716	0.65	578.41	877.36
3	0.48	556	0.23	268.30	14637.25	12	0.48	545	0.23	262.99	12096.59
4	0.17	217	0.03	35.90	731.11	13	0.17	199	0.03	32.92	81.71
5	0.00	11	0.00	0.05	3011.77	14	0.00	10	0.00	0.05	3122.53
6	0.08	60	0.01	4.56	3700.36	15	0.08	71	0.01	5.39	2483.09
7	0.35	341	0.12	117.81	141.57	16	0.35	351	0.12	121.27	479.53
8	0.69	651	0.47	447.43	3336.13	17	0.69	656	0.47	450.87	3938.72
9	0.94	740	0.89	696.69	2465.51	18	0.94	735	0.89	691.98	2987.05
–	–	–	–	–	–	Total	9.00	8073	6.75	5775.21	66372.37

4.1 Least square linear fitting for table 2 data

So the values of A and B are obtained for the data in table 6 as follows:

$$\begin{aligned}\Delta &= N\sum x_i^2 - \left(\sum x_i\right)^2 \\ &= (41 \times 10.83) - (14.84)^2 = 223.7\end{aligned}$$

$$
\begin{aligned}
A &= \frac{\sum x_i^2 \sum y_i - \sum x_i \sum x_i y_i}{\Delta} \\
&= \frac{(10.83 \times 9902\,\mathrm{mV}) - (14.84 \times 6011.20\,\mathrm{mV})}{223.8} \\
&= 80.51\,\mathrm{mV}
\end{aligned}
$$

$$
\begin{aligned}
B &= \frac{N \sum x_i y_i - \sum x_i \sum y_i}{\Delta} \\
&= \frac{(41 \times 6011.20\,\mathrm{mV}) - (14.84 \times 9902\,\mathrm{mV})}{223.8} \\
&= 444.9\,\mathrm{mV}
\end{aligned}
$$

To calculate the error in the parameters from least square linear fit, the calculations are performed as follows:

$$
\begin{aligned}
\sigma_y &= \sqrt{\frac{1}{N-2}\sum_{i=1}^{N}(y_i - A - Bx_i)^2} = \sqrt{\frac{1}{41-2}\sum_{i=1}^{41}(y_i - A - Bx_i)^2} \\
&= \sqrt{\frac{1}{39} \times 207215.06\,\mathrm{mV}^2} = 72.89\,\mathrm{mV}
\end{aligned}
$$

$$
\begin{aligned}
\sigma_A &= \sigma_y \sqrt{\frac{\sum x_i^2}{\Delta}} = 72.89\,\mathrm{mV} \times \sqrt{\frac{10.83}{223.8}} = 16.03\,\mathrm{mV} \\
\sigma_B &= \sigma_y \sqrt{\frac{N}{\Delta}} = 72.89\,\mathrm{mV} \times \sqrt{\frac{41}{223.8}} = 31.20\,\mathrm{mV}
\end{aligned}
$$

4.2 Least square linear fitting for table 5 data

The values of A and B are obtained for the data in table 6 as follows:

$$
\begin{aligned}
\Delta &= N \sum x_i^2 - \left(\sum x_i\right)^2 \\
&= (18 \times 6.75) - (9.00)^2 = 40.5
\end{aligned}
$$

$$
\begin{aligned}
A &= \frac{\sum x_i^2 \sum y_i - \sum x_i \sum x_i y_i}{\Delta} \\
&= \frac{(6.75 \times 8073\,\mathrm{mV}) - (9.00 \times 5775.21\,\mathrm{mV})}{40.5} \\
&= 62.1\,\mathrm{mV}
\end{aligned}
$$

$$
\begin{aligned}
B &= \frac{N \sum x_i y_i - \sum x_i \sum y_i}{\Delta} \\
&= \frac{(18 \times 5775.21\,\mathrm{mV}) - (9.00 \times 8073\,\mathrm{mV})}{40.5} \\
&= 772.8\,\mathrm{mV}
\end{aligned}
$$

To calculate the error in the parameters from least square linear fit, the calculations are performed as follows:

$$
\begin{aligned}
\sigma_y &= \sqrt{\frac{1}{N-2}\sum_{i=1}^{N}(y_i - A - Bx_i)^2} = \sqrt{\frac{1}{18-2}\sum_{i=1}^{18}(y_i - A - Bx_i)^2} \\
&= \sqrt{\frac{1}{16} \times 66372.37\,\mathrm{mV}^2} = 64\,\mathrm{mV}
\end{aligned}
$$

$$
\begin{aligned}
\sigma_A &= \sigma_y \sqrt{\frac{\sum x_i^2}{\Delta}} = 64\,\mathrm{mV} \times \sqrt{\frac{6.75}{40.5}} = 26\,\mathrm{mV} \\
\sigma_B &= \sigma_y \sqrt{\frac{N}{\Delta}} = 64\,\mathrm{mV} \times \sqrt{\frac{18}{40.5}} = 43\,\mathrm{mV}
\end{aligned}
$$

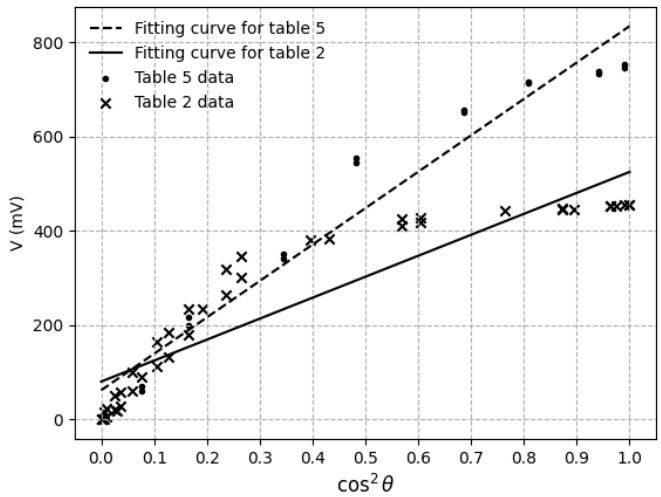


Figure 3: Malus's law plot for table 2 and table 5 with their least square linear fit curve.

5 Results and Discussion

The values of the parameters in the least square linear fit for the data in table 2 is as follows:

$$A = 81 \pm 16\,\text{mV}$$
$$B = 445 \pm 31\,\text{mV}$$

The values of the parameters in the least square linear fit for the data in table 5 is as follows:

$$A = 62 \pm 26\,\text{mV}$$
$$B = 773 \pm 43\,\text{mV}$$

The graph plot of the Malus's law curve in both table 2 and table 5 do not resemble a straight line as predicted by Malus's law. However, when comparing both the graphs, we see that for lower values of $\cos^2\theta$, the plots for both the graphs coincide and are linear. For larger values of $\cos^2\theta$ (larger than 0.3) we see the graph plots start to differ and show a saturation as $\cos^2\theta$. approaches 1. The experiment can conclude that Malus's law has been qualitatively verified.

In table 1, the voltage reading shows a maximum reading of 455 mV in 70° (corrected position: 1°) and 250° (corrected position: 181°) of the analyser while a minimum reading of 0 mV in 340° (corrected position: 271°) and 155° (corrected position: 86°) of the analyser. The minima are separated by 185° while the maxima are separated by 180°. The angle between a minima and maxima is close to 90°.

Table 3 is when an half-wave plate is positioned at 292° orientation whereas the polariser is placed at 62°. So, the axis of polarisation is 62°. Since the direction of polarisation has a ±180° ambiguity (electric field is oscillating) the direction of polarisation can be added by 180° to give 242°. So, the plane polarised light is oriented at an angle of $292° - 242° = 50°$ orientation to the fast axis of the half-wave plate. The maxima for the beam is observed at 340° (corrected position: 271°) and 160° (corrected position: 91°) of the analyser. Hence, after adding the half-wave plate at a 50° to the polariser, the plane of polarisation of the beam rotated by $340° - 250° = 90°$. We expected a rotation of $2\theta = 2 \times 50° = 100°$. The 10° discrepency might be due to the difference in the calibrated 0° of the half-wave plate and the fast axis of the wave-plate.

Table 4 is when a quarter-wave plate is positioned at 226° orientation. The minimum value of the voltage reading is 386 mV while a maximum in the voltage is 396 mV which suggests that the intensity is nearly constant in each direction. Although such an observation can also be made for an unpolarised light, given than we find the data to vary by 10 mV when turning the analyser by 360°, the light cannot be unpolarised where the voltage reading will not change at all. The maxima and minima are alternating in $\sim 90°$ rotation of the analyser which resembles the light to be elliptically polarised. Since circularly polarised light is also elliptically polarised light with the intensity showing very low variation with the analyser angle, we can conclude that the light obtained using the quarter-wave plate is circularly polarised light.

6 Conclusion

The experiment successfully fulfilled its objectives. The Malus's law is verified qualitatively. The plane polarised light was rotated by $\sim 90^\circ$ using the half-wave plate and circularly polarised light is obtained from plane polarised light using a quarter-wave plate.